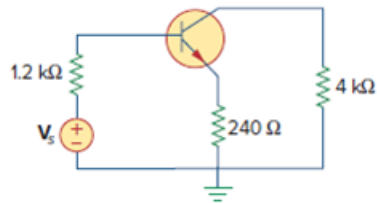


Problem

Determine A_v , A_i , Z_{in} , and Z_{out} for the amplifier shown in Fig. 19.131. Assume that

$$\begin{array}{ll} h_{ie} = 4 \text{ k}\Omega, & h_{re} = 10^{-4} \\ h_{fe} = 100, & h_{oe} = 30 \text{ }\mu\text{S} \end{array}$$

Figure 19.131



Step-by-step solution

Step 1 of 11

Consider the following circuit diagram:

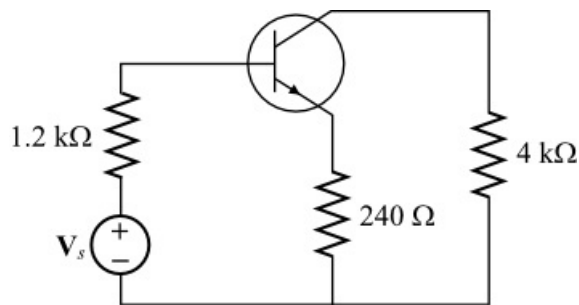


Figure 1

Step 2 of 11

Draw the equivalent circuit diagram of Figure 1.

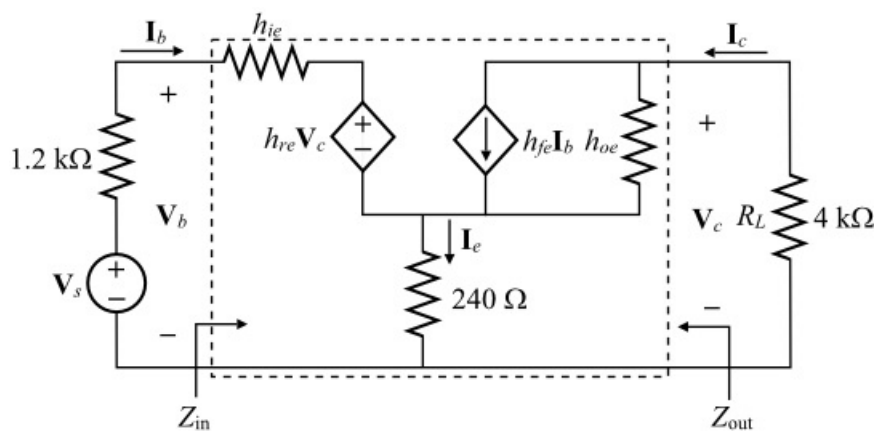


Figure 2

Step 3 of 11

From the figure 2, write the expression for $240\ \Omega$ using Kirchhoff's current law.

$$\mathbf{I_e = I_b + I_c \dots\dots (1)}$$

Apply Kirchhoff's voltage law to left side loop.

$$\mathbf{V_b = h_{ie}I_b + h_{re}V_c + 240I_e}$$

$$\mathbf{V_b = h_{ie}I_b + h_{re}V_c + 240(I_b + I_c) \dots\dots (2)}$$

Apply Kirchhoff's current law at node V_c .

$$\mathbf{I_c = h_{fe}I_b + \frac{V_c}{240 + \frac{1}{h_{oe}}} \dots\dots (3)}$$

Step 4 of 11

Write the expression for output voltage.

$$V_c = -I_c R_L$$

Substitute $-I_c R_L$ for V_c in the equation (3).

$$I_c = h_{fe} I_b + \frac{-I_c R_L}{240 + \frac{1}{h_{oe}}}$$

$$\left(1 + \frac{R_L}{240 + \frac{1}{h_{oe}}}\right) I_c = h_{fe} I_b$$

$$\left(\frac{240 + \frac{1}{h_{oe}} + R_L}{240 + \frac{1}{h_{oe}}}\right) I_c = h_{fe} I_b$$

$$\left(\frac{1 + h_{oe}(R_L + 240)}{1 + h_{oe} 240}\right) I_c = h_{fe} I_b$$

On further simplification:

$$\frac{I_c}{I_b} = \frac{h_{fe}(1 + h_{oe} 240)}{1 + h_{oe}(R_L + 240)} \dots\dots (4)$$

$$A_i = \frac{h_{fe}(1 + h_{oe} 240)}{1 + h_{oe}(R_L + 240)} \quad \left(\text{since } \frac{I_c}{I_b} = A_i\right)$$

$$= \frac{(100)[1 + (30 \times 10^{-6})(240)]}{1 + (30 \times 10^{-6})(4000 + 240)}$$

$$= \frac{(100)[1.0072]}{1 + 1272}$$

$$= \frac{100.72}{1.1272}$$

$$= 89.35$$

Therefore, the current gain of the amplifier, A_i is 89.35.

From the equation (4),

$$I_c = \left(\frac{h_{fe}(1 + h_{oe} 240)}{1 + h_{oe}(R_L + 240)}\right) I_b$$

Substitute $h_{fe} I_b + \frac{V_c}{240 + \frac{1}{h_{oe}}}$ for I_c in the above equation.

$$\left(\frac{h_{fe}(1 + h_{oe} 240)}{1 + h_{oe}(R_L + 240)}\right) I_b = h_{fe} I_b + \frac{V_c}{240 + \frac{1}{h_{oe}}}$$

$$\left(\frac{h_{fe}(1 + h_{oe} 240)}{1 + h_{oe}(R_L + 240)} - h_{fe}\right) I_b = \frac{V_c}{240 + \frac{1}{h_{oe}}}$$

$$I_b = \frac{V_c}{\left(240 + \frac{1}{h_{oe}}\right) \left(\frac{h_{fe}(1 + h_{oe} 240)}{1 + h_{oe}(R_L + 240)} - h_{fe}\right)}$$

Substitute $\frac{V_c}{\left(240 + \frac{1}{h_{oe}}\right)\left(\frac{h_{fe}(1+h_{oe}240)}{1+h_{oe}(R_L+240)} - h_{fe}\right)}$ for I_b and $-\frac{V_c}{R_L}$ for I_c in the equation (2).

$$\begin{aligned} V_b &= h_{ie}I_b + h_{re}V_c + 240(I_b + I_c) \\ &= (h_{ie} + 240) \left[\frac{V_c}{\left(240 + \frac{1}{h_{oe}}\right)\left(\frac{h_{fe}(1+h_{oe}240)}{1+h_{oe}(R_L+240)} - h_{fe}\right)} \right] + h_{re}V_c - \frac{240V_c}{R_L} \\ &= \left[\frac{h_{ie} + 240}{\left(240 + \frac{1}{h_{oe}}\right)\left(\frac{h_{fe}(1+h_{oe}240)}{1+h_{oe}(R_L+240)} - h_{fe}\right)} + h_{re} - \frac{240}{R_L} \right] V_c \\ \frac{V_b}{V_c} &= \left[\frac{h_{ie} + 240}{\left(240 + \frac{1}{h_{oe}}\right)\left(\frac{h_{fe}(1+h_{oe}240)}{1+h_{oe}(R_L+240)} - h_{fe}\right)} + h_{re} - \frac{240}{R_L} \right] \end{aligned}$$

Step 6 of 11

On Further simplification:

$$\begin{aligned} \frac{1}{A_v} &= \left[\frac{h_{ie} + 240}{\left(240 + \frac{1}{h_{oe}}\right)\left(\frac{h_{fe}(1+h_{oe}240)}{1+h_{oe}(R_L+240)} - h_{fe}\right)} + h_{re} - \frac{240}{R_L} \right] \\ &= \left[\frac{4000 + 240}{\left(240 + \frac{1}{30 \times 10^{-6}}\right)\left(\frac{(100)(1+(30 \times 10^{-6})(240))}{1+(30 \times 10^{-6})(4000+240)} - 100\right)} + 10^{-4} - \frac{240}{4000} \right] \\ &= \left[\frac{4240}{(33573.33)\left(\frac{(100)(1.0072)}{1.1272} - 100\right)} + 10^{-4} - 0.06 \right] \\ &= \left[\frac{4240}{(33573.33)(89.35 - 100)} + 10^{-4} - 0.06 \right] \\ &= [-0.0118 + 10^{-4} - 0.06] \\ &= -0.07176 \\ A_v &= \frac{1}{-0.07176} \\ &= -13.93 \end{aligned}$$

Step 7 of 11

Therefore, the voltage gain of the amplifier, A_v is -17.74.

Step 8 of 11

From the equation (4):

$$\frac{I_c}{I_b} = \frac{h_{fe}(1+h_{oe}240)}{1+h_{oe}(R_L+240)}$$

$$I_c = \frac{h_{fe}(1+h_{oe}240)}{1+h_{oe}(R_L+240)} I_b$$

Substitute $-I_c R_L$ for V_c in the equation (2).

$$V_b = h_{ie} I_b + h_{re} (-I_c R_L) + 240(I_b + I_c)$$

$$= (h_{ie} + 240) I_b + (240 - h_{re} R_L) I_c$$

Now substitute $\frac{h_{fe}(1+h_{oe}240)}{1+h_{oe}(R_L+240)} I_b$ for I_c in the above equation.

$$V_b = (h_{ie} + 240) I_b + (240 - h_{re} R_L) \left(\frac{h_{fe}(1+h_{oe}240)}{1+h_{oe}(R_L+240)} \right) I_b$$

$$\frac{V_b}{I_b} = h_{ie} + 240 + (240 - h_{re} R_L) \left(\frac{h_{fe}(1+h_{oe}240)}{1+h_{oe}(R_L+240)} \right)$$

$$Z_{in} = h_{ie} + 240 + \left(\frac{h_{fe}(1+h_{oe}240)(240 - h_{re} R_L)}{1+h_{oe}(R_L+240)} \right)$$

Calculate the value of the input impedance.

$$Z_{in} = h_{ie} + 240 + \left(\frac{h_{fe}(1+h_{oe}240)(240 - h_{re} R_L)}{1+h_{oe}(R_L+240)} \right)$$

$$= 4000 + 240 + \left(\frac{(100)(1 + (30 \times 10^{-6})(240))(240 - (10^{-4})(4000))}{1 + (30 \times 10^{-6})(4000 + 240)} \right)$$

$$= 4240 + \left(\frac{(100)(1.0072)(240 - 0.4)}{1.1272} \right)$$

$$= 4240 + \frac{(100)(1.0072)(239.6)}{1.1272}$$

$$= 4240 + 21409.25$$

$$= 25.65 \text{ k}\Omega$$

Therefore, the input impedance, Z_{in} is 25.65 kΩ.

Step 9 of 11

To calculate the output impedance, short the input source and replace load with 1 V voltage source. Draw the modified circuit diagram:

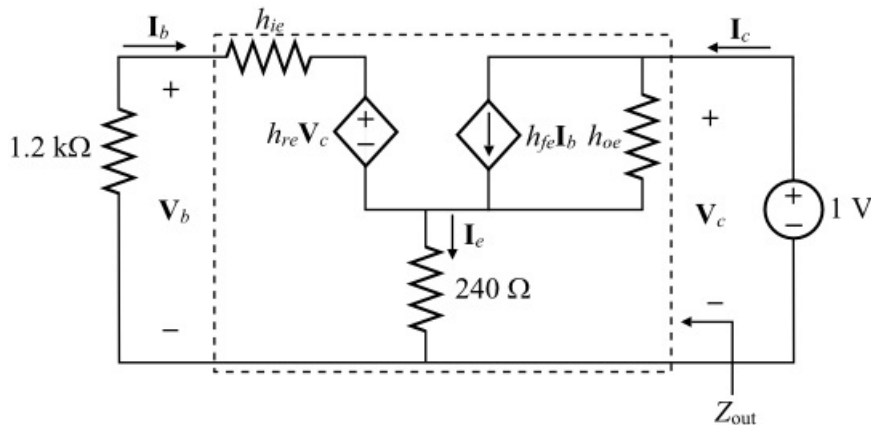


Figure 3

Step 10 of 11

Apply Kirchhoff's voltage law to left side loop.

$$(h_{ie} + 1200)I_b + h_{re}V_c + 240I_e = 0$$

But $V_c = 1 \text{ V}$

$$(h_{ie} + 1000)I_b + h_{re} + 200(I_b + I_c) = 0$$

$$(h_{ie} + 1200 + 240)I_b + h_{re} + 240I_c = 0 \dots\dots (5)$$

Apply Kirchhoff's current law at node V_c .

$$I_c = h_{fe}I_b + \frac{1}{240 + \frac{1}{h_{oe}}}$$

$$h_{fe}I_b = I_c - \frac{1}{240 + \frac{1}{h_{oe}}}$$

$$I_b = \frac{1}{h_{fe}} \left(I_c - \frac{1}{240 + \frac{1}{h_{oe}}} \right) \dots\dots (6)$$

Substitute $\frac{1}{h_{fe}} \left(\mathbf{I}_c - \frac{1}{240 + \frac{1}{h_{oe}}} \right)$ for $\mathbf{I}_b$ in the equation (5).

$$(h_{ie} + 1200 + 240) \left(\frac{1}{h_{fe}} \left(\mathbf{I}_c - \frac{1}{240 + \frac{1}{h_{oe}}} \right) \right) + h_{re} + 240\mathbf{I}_c = 0$$

$$\left(\frac{h_{ie} + 1200 + 240}{h_{fe}} + 240 \right) \mathbf{I}_c = \frac{h_{ie} + 1200 + 240}{h_{fe} \left(240 + \frac{1}{h_{oe}} \right)} - h_{re}$$

$$\mathbf{I}_c = \left(\frac{h_{fe}}{h_{ie} + 1440 + 240h_{fe}} \right) \left(\frac{h_{oe}(h_{ie} + 1440)}{h_{fe}(1 + 240h_{oe})} - h_{re} \right)$$

Calculate the output impedance.

$$\begin{aligned} Z_{out} &= \frac{1}{\mathbf{I}_c} \\ &= \frac{1}{\left(\frac{h_{fe}}{h_{ie} + 1440 + 240h_{fe}} \right) \left(\frac{h_{oe}(h_{ie} + 1440)}{h_{fe}(1 + 240h_{oe})} - h_{re} \right)} \\ &= \frac{4000 + 1440 + (240)(100)}{(100) \left(\frac{(30 \times 10^{-6})(4000 + 1440)}{(100)(1 + (240)(30 \times 10^{-6}))} - 10^{-4} \right)} \\ &= \frac{5440 + 24000}{(100) \left(\frac{0.1632}{100.72} - 10^{-4} \right)} \\ &= \frac{29440}{(100)(1.5203 \times 10^{-3})} \\ &= \frac{29440}{0.152} \\ &= 193.7 \text{ k}\Omega \end{aligned}$$

Therefore, the output impedance, Z_{out} is 193.7 kΩ.